
Part VI

General Medical Considerations

Anorexia Athletica and Relative Energy Deficiency

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16.1 Introduction

Several sporting disciplines have a recognized benefit from low body weight, namely, sports with weight categories like martial arts and sports in which the body weight needs to be moved against gravity like gymnastics and of course sport climbing. As a consequence, some climbers control their weight through at least one unhealthy way [1]. This is called relative energy deficiency in sports or RED-S and can have serious long-term complications: dental caries, osteoporosis, renal insufficiency, anemia, and also psychiatric conditions. Even though RED-S, formerly known as anorexia athletica, is not recognized as a true disorder, it can lead to psychiatric disorders like anorexia nervosa or depression. Some of the athletes are so convinced of the dependence of their weight on their performance, and this wrong belief is so firmly ingrained that they suffer their whole life from the mental consequences.

So far there are no restrictions on athletes diagnosed with RED-S by the IFSC, as long as it is medically justifiable, especially as restricting the athlete could lead to a serious aggravation of the athlete's situation. Instead of receiving help from his or her federation, the athlete could become isolated increasing their risk of serious disorders following in the wake of RED-S. However, a regular monitoring of BMI is being conducted during world cup competitions. If concerns are raised, the IFSC can oblige the athlete to get a medical check-up [2]. An early detection of possible warning signs and support for the athlete must be a high priority in order to avoid long-term damage.

16.2 RED-S (Relative Energy Deficiency in Sport)

Relative energy deficiency in sport (RED-S) describes a physiology as a consequence of relative energy deficiency and includes limitations to the metabolism, the menstrual cycle, the bone health, the immune system, the protein synthesis, and the cardiopulmonary function [3]. In theory, every athlete is at an increased risk of RED-

S. However, some sports present a higher risk factor. These are sports with an aspect of aesthetics and appearance, weight-class sports, endurance sports [3], and sports in which the own body weight needs to be moved against gravity (climbing, gymnastics). In 2014, the IOC published a "consensus statement" in which all the signs and risks of RED-S were explained [4]. One year later, a diagnostic tool was developed, the RED-S CAT [3].

16.3 RED-S in Climbing

An optimal balance between low body weight, or a low body fat percentage, respectively, with still a high amount of strength plays an essential role in climbing [5]. This is a consequence of the fact that the own body weight needs to be propelled against gravity often with only the fingers doing the whole work. The endurance aspect in climbing is rather low compared to endurance sports but probably comparable to team sports or martial arts [6, 7]. Training, competition, and even climbing outdoors most likely resemble interval training [8]. Restricting the diet and reducing especially the amount of incorporated fatty acids in order to keep the body fat low is therefore a regular occurrence in climbers as the need for a sufficient amount of nutrients is not felt as much as in endurance sports. Another option is using an endurance sport for losing weight and body fat, a method employed by many climbers excessively.

16.3.1 Energy and Nutrient Supply in Sport Climbers

Male elite climbers have a lower BMI than elite athletes from other sports, even when compared to long-distance runners [1, 9, 10]. The mean energy supply in climbers is comparable to elite athletes from other sports in which the body weight needs to be low (gymnastics, dancing) or sports with weight classes (wrestling, martial arts) and is decidedly lower than in endurance athletes [1]. However, endurance athletes also have a much higher energy demand as a con-

sequence of their specific training regimes, and therefore a direct comparison is not feasible. In addition to that, it is not only essential how much food is consumed but its components. A study on climbers nutritional intake was able to show that a small part comes from visible fat or meat (12% of the climbers vs. 60% general pediatric population [1]). This is less than in any other sport discipline [1]. Especially the female elite climbers tend to restrict their calories with a very low protein content leading to very low protein and caloric levels in their blood [1]. Furthermore, the composition of the food is mainly inappropriate. The combination of little food intake with high training volumes leads to malnutrition on the one hand and sudden ravenousness which can only be satisfied with fast food low on nutrients [1]. This results in a shortage of micronutrients.

16.3.2 Correlations Between Climbing Performance and BMI/Body Fat

The influence of a low body weight on the climbing performance needs to be questioned critically. Several studies have shown that good climbers tend to be exceptionally small and thin [1, 5, 11, 12]. However, it is not possible to draw the conclusion that there exists a causality between body weight and climbing level. It also needs to be stressed that especially the mental belief that a low body weight can influence the climber's performance is so deeply embedded in some climbers that it becomes a self-fulfilling prophecy. Even though this may seem harmless at first, the energy and nutrient restrictions following such beliefs can cause long-term impairment of health and performance and may therefore limit the performance of the athlete in the long term.

16.3.3 Eating Disorders in Climbers

The belief that a low body weight represents the key to success in climbing is widely spread

among climbers. However, are this belief and the actions taken in order to reach certain weight goals already considered a disorder?

The full scale of anorexia nervosa or bulimia nervosa is a recognized psychiatric disorder with clear diagnostic criteria. Climbers have many characteristics in common with people who have a tendency toward eating disorder. They are perfectionists, reliant, and ready to work hard, have a high need for approval, can tolerate pain and discomfort well, and plan their eating manners according to the demands of their sport [1]. On the other hand, there are some characteristics that do not fit: low self-esteem coupled with being discontent with one own performance; a distinct distorted perception of the own body coupled with a fear of gaining weight; a rigid eating and training regimen, which is kept up during season breaks. These characteristics lead to athletes who can have successful careers in spite of their disorders and never acknowledge their disease [1, 13]. In some cases, athletes have even taken laxatives and diuretics in order to lose weight [13].

RED-S is observed more frequently in women than in men and occurs mainly between the ages of 15–25. However, we've had male athletes who were affected severely.

The consequences of restricting one's diet can be severe: amenorrhea, stress fractures, osteoporosis, changes in metabolism, kidney and liver conditions, depression, and other psychiatric diseases. In many cases, the self-restricting diet leads to permanent damage as has been reported by some athletes openly. Angela Eiter, former world champion in lead climbing reported openly about her ongoing problems with the topic in a European Newspaper (Neue Zürcher Zeitung). After having been asked to lose some weight in order to be more performant in climbing, she was more successful, and this has impacted on her entire eating habits to this day.

Since a low body weight is considered normal in climbing, a pronounced vigilance is necessary from coaches, physicians, and peers in order to detect early warning signs.

16.4 Definition RED-S CAT and Its Implementation

The RED-S CAT is a clinical assessment tool for the evaluation of athletes/active individuals suspected of having relative energy deficiency [3]. It is meant for use by medical professionals in the clinical evaluation and management of athletes with this syndrome [3]. The goal is to screen suspected athletes according to this tool and then place them in one of the categories. It is based on the IOC Consensus Statement on RED-S [4].

16.4.1 Red Light

In the red light, we find athletes with one of the following diagnosed disorders:

- Anorexia nervosa and other eating disorders
- Other serious medical (psychological and physiological) conditions related to low energy availability
- Use of extreme weight loss techniques leading to dehydration-induced hemodynamic instability and other life-threatening conditions
- Severe ECG abnormalities (i.e., bradycardia)

Athletes in the red light need medical help, and due to the seriousness of the presentation, sport participation represents a serious threat. It may also distract the athlete from devoting the attention needed for treatment and recovery. Athletes in this category should be forbidden to participate in competitions or serious training [3].

16.4.2 Yellow Light

Here, we find athletes with a moderate risk which can be identified with the following characteristics:

- Prolonged abnormally low body fat (measured using DXA or anthropometric measurements)
- Substantial weight loss of 5–10% in 1 month
- Attenuation of expected growth and development in adolescents

- Low energy intake over a longer time period
- Abnormal menstrual cycle with amenorrhea over more than 3 months
- No menarche at 15 years of age
- Low mineral bone density (DXA measurements)
- Stress fractures
- Psychological complications from low energy intake
- Disordered eating behavior negatively affecting other team members
- Lack of progress in treatment and/or noncompliance

Athletes in the yellow light are allowed to participate in competitions if medically supervised and treated. Reevaluation of the athlete's risk assessment should occur at regular intervals of 1–3 months depending on the clinical scenario to assess compliance and to detect changes in clinical status [3].

16.4.3 Green Light

Athletes in the green light only have a low risk. They can be classified according to the following criteria:

- Appropriate physique that is managed without unhealthy diet/exercise strategies
- Healthy eating habits
- Healthy endocrine system
- Healthy bone mineral density
- Healthy musculoskeletal system

There are no restrictions for these athletes [3].

16.4.4 Treatment

Athlete in the red light should therefore not participate in competitions or training. They need to sign a written contract in which the therapy goals are spelled out so that they know when a return to sport is possible. Athletes in the yellow light can participate in competitions but should be monitored closely and follow an exact treatment plan. Athlete in the green light can continue as before [3].

This represents also the approach already practiced by the IFSC. In order to monitor the developments in the IFSC athletes, BMI measurements for screening during world cup competitions have become routine. If an abnormality during these routine screenings is observed, the national federation is informed and is asked to take appropriate steps in order to categorize the athlete and then provide appropriate treatment plans if indicated. It is also up to the national federation if the athlete needs to leave the competition circuit or not. If the athlete is below a minimum BMI the athlete is only granted an international permit after the federation's physician has provided a written statement allowing the athlete to keep competing [2]. However, this concept is in perpetual evolvement.

16.5 Conclusion

The health of each athlete must be the priority for each physician working with climbers. If an eating disorder or relative energy deficiency is suspected, this needs to be addressed promptly. There can be no beating about the bush since the consequences can be permanent. Especially, adolescents can experience worsening of a little energy restriction into severe psychiatric disorders from which they may never recover. The RED-S CAT represents a useful tool for detecting and judging the extent of an eating disorder in athletes and provides guidelines for helping these athletes.

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